Basmala Nassar

SOFTWARE ENGINEERING · B.SC. COMPUTER ENGINEERING · TECHNION

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Final-year Computer & Software Engineering student at the Technion who builds **systems software in C, C++ and Python**. Shipped six substantial projects — four computer-architecture simulators, a Linux shared library, and a layered network-stack model — each driven to a **fully green automated test suite** and Valgrind-clean. Strong grounding in **data structures, algorithms, operating systems, computer networks and formal verification**, with daily **Linux** fluency and comfort debugging from application code down to the binary.

> **Expected graduation:** October 2027 (penultimate/final year) · **Internship availability:** from March 2027, 2–4 days/week for 26+ weeks · **Location:** Haifa / Tel Aviv, Israel · no sponsorship required

EDUCATION

B.Sc. Computer and Software Engineering, Technion
Nov 2023 – Oct 2027 (expected)

Andrew & Erna Viterbi Faculty of Electrical and Computer Engineering, Haifa · GPA 81.6

- **Software & CS grades:** Basic Software Lab (98) · Software Systems, C/C++ (82) · Digital Systems & Computer Structure (82) · Introduction to CS, C (81) · Operating Systems (77)
- **Core coursework:** Data Structures & Algorithms · Operating Systems Structure · Computer Architecture · Computer Networks (TCP/IP) · Software Systems
- **Currently taking:** Correctness Guarantees for Software (SMT-based verification, abstract interpretation) · Dynamic Binary Translation & Optimization

PROJECTS

Computer Architecture Simulators — Four Systems Projects
C · C++ · Python · Linux

Computer Architecture (046267) · each graded against a hidden automated test suite

- **Branch-predictor simulator** — parameterized BTB supporting local/global history with GShare/LShare XOR indexing into per-branch or shared FSM tables; **458/458 tests passing**
- **Cache-hierarchy simulator** — two-level L1/L2 cache with LRU replacement, write-back / write-allocate policies and average-access-time statistics; **803/803 tests passing**
- **Dataflow dependency analyzer** — builds an instruction-dependency **DAG** and computes critical-path depths to model out-of-order execution; **500/500 tests, Valgrind-clean**
- **Dynamic binary profiler** — Intel Pin tool instrumenting real x86-64 ELF binaries in probe mode to collect and rank hot branch/loop profiles; root-caused a **SIGSEGV** on SPEC `sgcc` to a rewritten lazily-bound **PLT stub** and fixed it by excluding the `.plt` region, isolated under **GDB**
- Wrote the surrounding **Python** tooling: generated test inputs, diffed output against reference traces and summarised pass/fail across hundreds of cases per run

Linux Shared Library over an Opaque ADT
C · Make · Linux

Introduction to Software Systems (044101)

- Built **libgrades.so** from a hand-written Makefile (`-fPIC`, `-shared`), dynamically linked against a second shared object at load time
- Drove a linked-list ADT through an **opaque iterator** and **function-pointer callbacks** (clone/destroy), managing nested ownership of dynamically allocated strings **leak-free under Valgrind**

Layered Network Interface Card (NIC) Model
C++ · OOP

Introduction to Software Systems (044101)

- Implemented the physical, data-link and network layers (L1–L3) of the network stack as separate collaborating classes
- Designed the inter-layer interfaces so each could be tested in isolation, handling packet encapsulation and decapsulation

Additional Projects
Python · SystemVerilog

- **Numerical PDE solver** (Python · NumPy · Matplotlib) — implemented finite-difference schemes and visualized convergence
- **Pacman on FPGA**, EE Lab 1A (grade 90) — VGA controller, sprite rendering and FSM game logic in SystemVerilog; verified module-by-module in ModelSim, then synthesized and debugged on real hardware
- **Configurable systolic-array convolution IP** — year-long VLSI project starting 2026/27: parametric PE grid for convolution, verified against a Python/NumPy golden model

SKILLS

Languages	**C**, **C++**, **Python** (NumPy, Matplotlib), SystemVerilog, MATLAB · comfortable picking up new languages as a project requires
CS foundations	Data structures & algorithms, complexity analysis, object-oriented design, operating systems (processes, memory, concurrency), computer networks (TCP/IP), formal verification & SMT-based reasoning
Systems	Linux & command-line workflows, memory management, shared libraries & dynamic linking, x86-64 assembly & ELF, computer architecture (caches, branch prediction, pipelines)
Tools	Git, GCC/Make, GDB, Valgrind, Intel Pin, ModelSim, Quartus, Vivado
Practices	Automated testing & regression triage, root-cause debugging, golden-model comparison, scripting to automate repetitive work
Spoken	Arabic (native) · Hebrew (fluent) · English (fluent)

TEACHING & LEADERSHIP

- **Private physics tutor** (self-employed, 2023–present) — teach mechanics, E&M and waves to high-school and first-year university students; build custom problem sets per student
- **Mentor, Technion Equal Opportunities Program** — guided first-year students through course registration and study strategy